1 What must all physical quantities have?

A a direction and a magnitude

B a direction and a unit

C a magnitude and a prefix

D a magnitude and a unit

2 What is $0.25\,\mathrm{kN\,mm^{-2}}$ expressed in $\mathrm{N\,m^{-2}}$?

A $0.00025\,\mathrm{N\,m^{-2}}$

B $0.25\,\mathrm{N\,m^{-2}}$

C $250\,000\,\mathrm{N\,m^{-2}}$

D $250\,000\,000\,\mathrm{N\,m^{-2}}$

3 A student calculates the density of a solid steel cube in an experiment.

The measured mass is $975\,\mathrm{g} \pm 10\,\mathrm{g}$ and the measured length of side is $50\,\mathrm{mm} \pm 1\,\mathrm{mm}$.

What is the density of the steel?

A $7.8\,\mathrm{g\,cm^{-3}} \pm 3.0\%$

B $7.8\,\mathrm{g\,cm^{-3}} \pm 7.0\%$

C $7.8\,\mathrm{g\,cm^{-3}} \pm 11\%$

D $7.8\,\mathrm{g\,cm^{-3}} \pm 13\%$

4 The time period T of a pendulum is given by

$$T = 2\pi\left(\frac{L}{g}\right)^n$$

where L is the length of the pendulum and g is the acceleration of free fall.

The equation is homogeneous.

What is the value of n?

A -2 **B** $-\frac{1}{2}$ **C** $\frac{1}{2}$ **D** 2